HUNTER

ENGINEERING CIVIL AND SURVEY

Final Basis of Design Report
Waste Water
For
Finish Line Auto Club – Perimeter Center
Princess Drive and Loop 101
Scottsdale, Arizona 85255

COS Plan Check #: 1059-PA-2024
COS Case #: 19-ZN-2004#2

Registered Professional Engineer (Civil)
CERTIFICATE NO.
48630
BRIAN K.
SCHILTZ
Date Signed 8/11/25
ARIZONA U.S.A.

August 2025

Prepared by:
Brian Schiltz
Hunter Engineering, Inc.
10446 N. 74th Street. Suite 140
Scottsdale, AZ 85258

H.E. PROJECT NO. LGEC332

Table of Contents

Appendix

1.0 Introduction

This design report has been prepared under a contract from LGE Corporation, developer of the Finish Line Auto Club project. The purpose of this report is to provide a sewer analysis, required by the City of Scottsdale, to support this development. Preparation of this report has been done according to the procedures detailed in Chapter 7 of the City of Scottsdale's *Design Standards and Policy Manual* (Reference 1).

This development project is located along the west side of N. Perimeter Drive and along the east side of N. Pacesetter Way, between E. Pacesetter Way and E. Princess Drive, within the City of Scottsdale, Maricopa County, Arizona. The site is specifically located in the south half of Section 36, Township 4 North, Range 4 East of the Gila and Salt River Base and Meridian. Figure 1, in Appendix A, illustrates the location of the project site in relation to the City of Scottsdale street system. Access to the site is provided from N. Perimeter Drive and N. Pacesetter Way.

The proposed project site is currently a developed parcel within the Perimeter Center master development. The existing parcel is bound by existing commercial developments to the north and south, N. Perimeter Drive to the east and N. Pacesetter Way to the west.

The development proposes the construction of five new commercial buildings and one new industrial building. Site improvements will include construction of driveway entrances, parking lots, sidewalk/hardscape, landscape areas, and supporting infrastructure including new storm water drainage systems, water, sewer and fire line services. The overall project site is approximately 10.394± acres.

2.0 Existing Condition

The existing site consists of a single commercial building and associated parking lots with parking canopies. The site is part of the overall Scottsdale Perimeter Center master development. There are existing 8-inch VCP public sewer mains within N. Perimeter Drive and N. Pacesetter Way adjacent to the site. There are four existing 8-inch sewer services to the site, two off N. Perimeter Drive and two off N. Pacesetter Way.

The City of Scottsdale Water and Wastewater Department requested that a downstream sewer manhole be flow tested to determine the available capacity. The manhole tested (MH1) is located at East Bell Road east of North Hayden Road. The flow test was performed by RDH Environmental Services, LLC between October 31st, 2024 and November 11th, 2024. See Appendix C for the sewer flow test results.

MH1 near Bell and Hayden resulted in a highest measured maximum flow during the test period of 184.97 gpm. The main tested is a 15-inch PVC pipe. The capacity for this main is 1,152 gpm at a minimum 0.225% slope and a d/D ratio of 0.70. This existing main is at approximately 18% capacity with approximately 855 gpm of available flow. It should be noted that the city quarter section maps show this existing main as 12-inch, however the main was measures as 15-inch in the field. Those measurements are shown in the flow test report in Appendix C.

3.0 Proposed Sanitary Sewer System

This project proposes to connect to the two existing 8-inch sewer services off N. Perimeter Drive and one of the existing 8-inch sewer services off N. Pacesetter Way. The new private sanitary sewer system will consist of 6-inch sewer service lines to each proposed building and a new 8-inch private sewer main with three new manholes. The 6-inch service lines for Buildings A, B, C, and D will connect to the 8-inch private main and drain to the existing southeastern 8-inch sewer service to the site off N. Perimeter Drive. The 6-inch service line for Building E will connect to a proposed manhole and drain to the existing northeastern 8-inch sewer service to the site off N. Perimeter Drive. The 6-inch service line for the industrial building will connect to a proposed manhole and drain to the existing southwestern 8-inch sewer service to the site off N. Pacesetter Way. The Conceptual Utility Plan, in Appendix A, depicts the proposed sanitary sewer improvements.

4.0 Projected System Demands

Per Figure 7-1.2 Average Day Sewer Demand in the City of Scottsdale *Design Standards & Policies Manual* a demand of 0.5 gallons per day per building square foot was used for the five proposed commercial buildings. The design peaking factor is 3.0. See Appendix B for a summary of these calculations.

Since the City of Scottsdale does not have an industrial sewer demand or peaking factor, wastewater flows for the proposed industrial building were calculated in accordance with the CSDSPM (Reference 1) and City of Phoenix Water Services Department, Design Standards Manual for Water and Wastewater Systems, 2021. See Appendix C. This results in wastewater flows of 50 gpd per every 1,000 sf of building. The peaking factor was calculated using Harmon's Formula per the City of Phoenix design standards. A peaking factor analysis was conducted using two different population estimations. An average provided by the architect assumes a population of one person per 2,100 sf of building and results in a peaking factor of 4.37. The City of Scottsdale zoning parking requirements for manufacturing and industrial use assumes a population of one person per 500 sf of building and results in a peaking factor of 4.24. The greater peaking factor, 4.37, was utilized to determine peak flow. See Appendix B for a summary of these calculations.

According to the calculations provided in Appendix B the proposed site will have an estimated Average Daily Flow total of 45.3 gpm and a Peak Hour Flow of 138.4 gpm. Each building will have its own service with the highest building peak flow of 65.7 gpm. Therefore, the proposed sewer service is calculated based on the minimum slope of 1.20% for a 6-inch sewer service for each building.

The capacity of the proposed 6-inch sewer service lines at the minimum slope is 209 gpm on a max d/D ratio of 0.65. This capacity is greater than the Peak Hour Flow of 65.7 gpm for the largest building. Therefore, the max individual building flow will be less than the service capacity flows.

The existing 8-inch sewer service stubs to the site were each installed with a slope of 0.34%. Therefore, the capacity of the existing sewer services to the site were calculated based on the minimum slope of 0.34% for an 8-inch sewer line. The existing 8-inch sewer service that connects to the proposed 8-inch private sewer main will drain the highest peak flow totaling 111.3 gpm from Buildings A, B, C, and D. The capacity of the proposed 8-inch private sewer main with a minimum slope of 0.34% is 239 gpm on a max d/D ratio of 0.65. This capacity is greater than the total Peak Hour Flow of 111.3 gpm.

The peak hour flow of 138.4 gpm for this development is well below the flow tested available capacity of 855 gpm in the 15-inch main in Bell Road.

The sanitary sewer pipe and fitting material for this project has been designated as PVC SDR-35. Trenching and bedding details for this project are to be per MAG Standard Specifications Section 601. Trench width above the installed pipe may be as wide as necessary to properly brace/install the work. Bedding backfill and compaction shall be installed per MAG Standard Specification 601.4. Service lines should connect to sewer according to MAG Standard Detail No. 440-3.

5.0 Conclusion

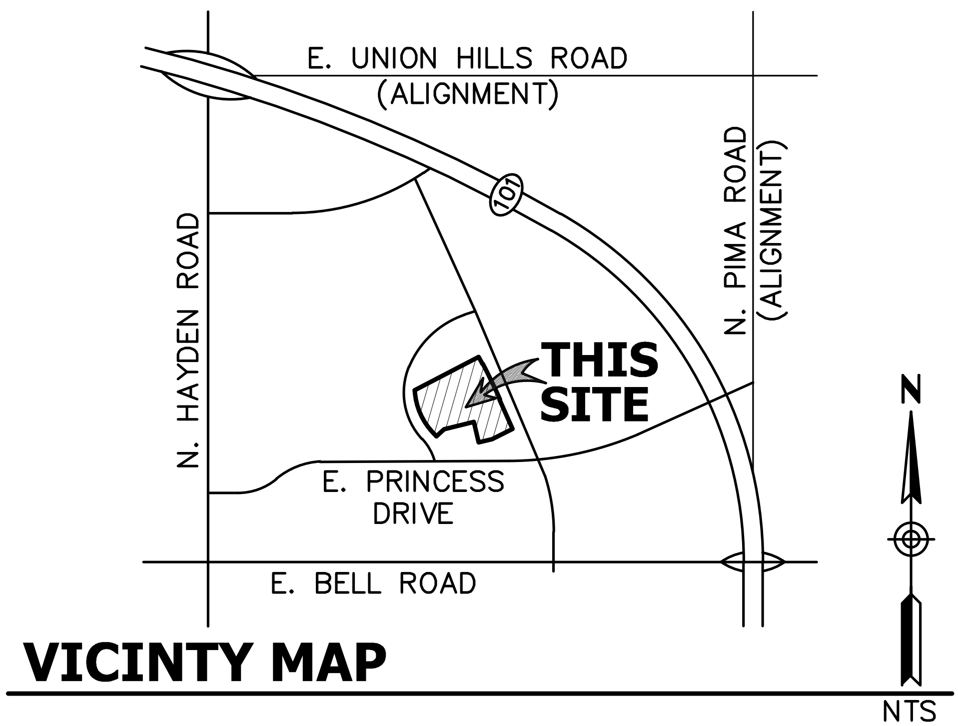
Based on the results of this study, it can be concluded that the proposed sanitary sewer system is adequate to support this development.

6.0 References

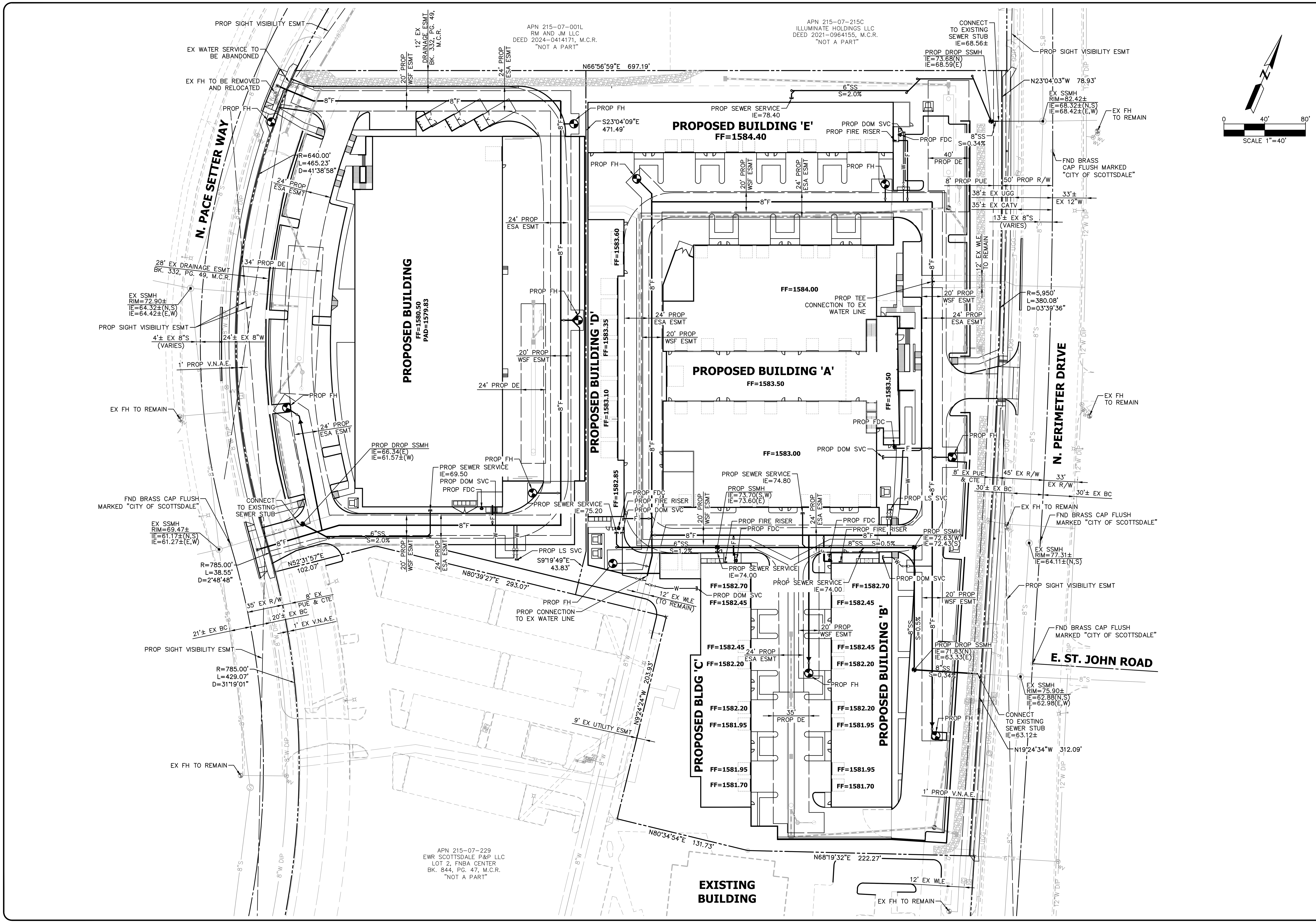
1) City of Scottsdale Design Standard & Policies Manual, January 2018 (Ref 1).

Appendix A

Vicinity Map and Conceptual Utility Plan



VICINITY MAP
FIGURE 1



NO.	DATE	REVISION	BY

PURPOSE: 3RD ZONING SUBMITTAL

DESIGN BY: BKS
DRAWN BY: MM
CHECKED BY: BKS

HUNTER ENGINEERING CIVIL AND SURVEY
10446 NORTH 74TH STREET SUITE 140
SCOTTSDALE, AZ 85258
P 480 991 3985

Registered Professional Engineer
CERTIFICATE NO. 48630
BRIAN K. SCHILTZ
ARIZONA, U.S.A.
Date Signed 8/11/25

CONCEPTUAL UTILITY PLAN FOR FINISH LINE AUTO CLUB PRINCESS DRIVE AND LOOP 101 SCOTTSDALE, ARIZONA

CONTACT ARIZONA 811 AT LEAST 2 FULL WORKING DAYS BEFORE YOU BEGIN EXCAVATION
ARIZONA811
CALL 811 OR CLICK ARIZONA811.COM

THESE PLANS ARE NOT APPROVED FOR CONSTRUCTION WITHOUT AN APPROVED SIGNATURE FROM THE GOVERNING MUNICIPALITY.

PROJECT NAME: FINISH LINE AUTO CLUB

HE NO.: LGEC332
SCALE: 1"=40'

SHEET: C4

Appendix B

Design Data and Calculations

Project: Finish Line Auto Club - Perimeter Center
Project No.: LGEC332
City: Scottsdale
Date: 7/17/2025

PROJECTED SANITARY SEWER LOADS

Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) Per Table 7-1.2 Average Day Sewer Demand	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor	Peak Flow (gpm)
Building A	Commercial	63,020	0.5 gpd/sf	31,510	21.9	3.00	65.7
Building B	Commercial	17,877	0.5 gpd/sf	8,939	6.2	3.00	18.6
Building C	Commercial	17,337	0.5 gpd/sf	8,669	6.0	3.00	18.0
Building D	Commercial	8,712	0.5 gpd/sf	4,356	3.0	3.00	9.0
Subtotal		**106,946**		**53,473**	**37.1**		**111.3**

Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) Per Table 7-1.2 Average Day Sewer Demand	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor	Peak Flow (gpm)
Building E	Commercial	18,558	0.5 gpd/sf	9,279	6.4	3.00	19.2
Subtotal		**18,558**		**9,279**	**6.4**		**19.2**

Total (Perimeter Drive)	**125,504**		**62,752**	**43.5**		**130.5**

Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) City of Phoenix Water Services Dept. Deisgn Standards Manual Table 8. Water and Wastewater Design Flows	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor Avg provided by Arch: one person per 2,100 sf of building	Peak Flow (gpm)	Peaking Factor COS Zoning Parking Requirement: one person per 500 sf of building	Peak Flow (gpm)
Industrial Building	Industrial	51,866	50.0 gpd/1000sf	2,593	1.8	4.37	7.9	4.24	7.6
Total (Pacesetter Way)		**51,866**		**2,593**	**1.8**		**7.9**		

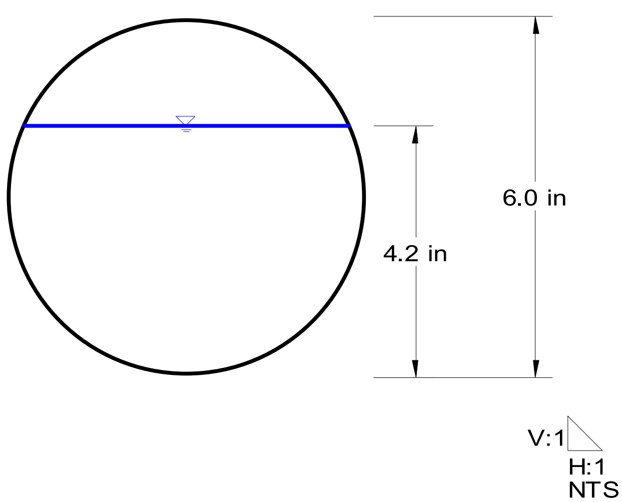
Grand Total	**177,370**		**65,345**	**45.3**		**138.4**

6-inch SS d/D=0.70 S=1.20%
Worksheet for Circular Channel

Project Description	
Worksheet	6-inch SS d/D=0.70 S=1.20%
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data		
Mannings Coefficient	0.013	
Channel Slope	1.20	%
Depth	4.2	in
Diameter	6.0	in

Results		
Discharge	231	gpm
Flow Area	0.1	ft²
Wetted Perimeter	0.99	ft
Top Width	0.00	ft
Critical Depth	0.37	ft
Percent Full	70.0	%
Critical Slope	1.07	%
Velocity	3.51	ft/s
Velocity Head	0.19	ft
Specific Energy	6.5	in
Froude Number	1.09	
Maximum Discharge	297	gpm
Discharge Full	276	gpm
Slope Full	0.84	%
Flow Type	Supercritical	

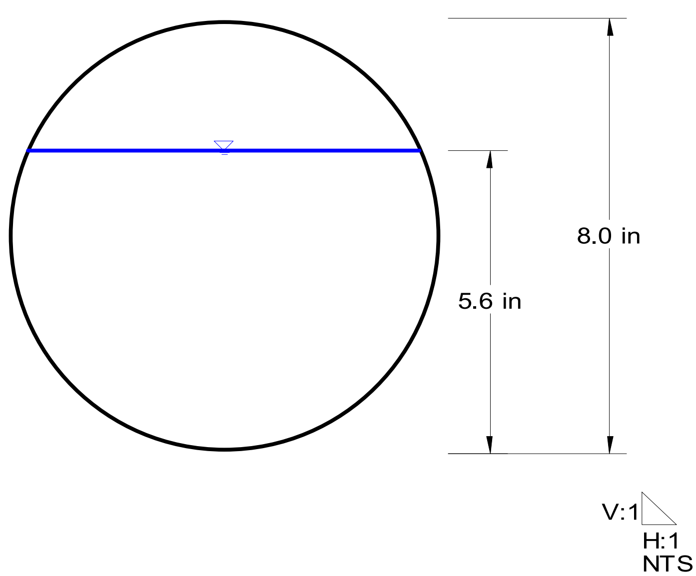


8-inch SS d/D=0.70 S=0.34%
Worksheet for Circular Channel

Project Description	
Worksheet	8-inch SS d/D=0.70 S=0.34%
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data		
Mannings Coefficient	0.013	
Channel Slope	0.34	%
Depth	5.6	in
Diameter	8.0	in

Results		
Discharge	265	gpm
Flow Area	0.3	ft²
Wetted Perimeter	1.32	ft
Top Width	0.00	ft
Critical Depth	0.36	ft
Percent Full	70.0	%
Critical Slope	0.73	%
Velocity	2.26	ft/s
Velocity Head	0.08	ft
Specific Energy	6.6	in
Froude Number	0.61	
Maximum Discharge	340	gpm
Discharge Full	316	gpm
Slope Full	0.24	%
Flow Type	Subcritical	

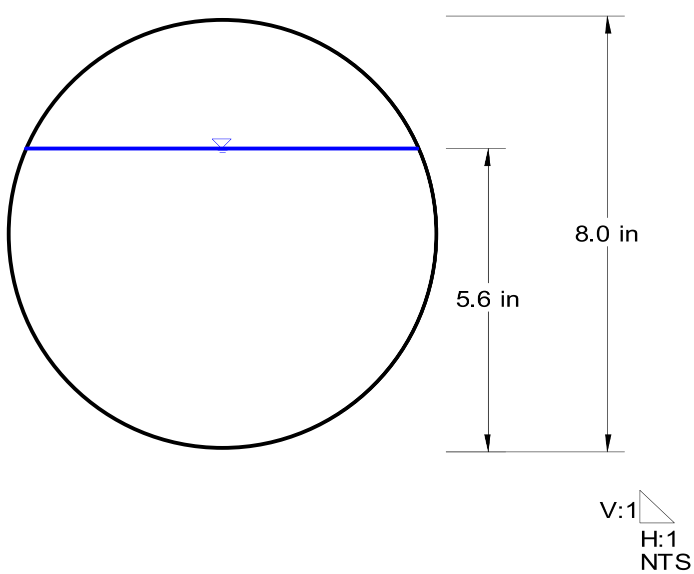


8-inch SS d/D=0.70 S=0.50%
Worksheet for Circular Channel

Project Description	
Worksheet	8-inch SS d/D=0.70 S=0.50%
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data		
Mannings Coefficient	0.013	
Channel Slope	0.50	%
Depth	5.6	in
Diameter	8.0	in

Results		
Discharge	321	gpm
Flow Area	0.3	ft²
Wetted Perimeter	1.32	ft
Top Width	0.00	ft
Critical Depth	0.40	ft
Percent Full	70.0	%
Critical Slope	0.78	%
Velocity	2.74	ft/s
Velocity Head	0.12	ft
Specific Energy	7.0	in
Froude Number	0.74	
Maximum Discharge	413	gpm
Discharge Full	383	gpm
Slope Full	0.35	%
Flow Type	Subcritical	



Appendix C

Reference Information

SL1780 RDH Flow Study for LGE

Blake Wells
LGE Design Build
1200 N. 52nd St, Phoenix, AZ 85008

SL1780 RDH Flow Study, 2 sites total in Scottsdale, AZ from Thursday 10-31-24 to Monday 11-11-24.

Equipment for Site: Hach 901 Logger with Flo-Dar Sensor (Area Velocity)**.**
The equipment was installed on Thursday, 10/31/24 with confined space entry, pipe size confirmed, sensor calibrated, and level depth confirmed to the flow level.

Duration of monitoring: 9 days
Monitor: Flow (gpm), Level (in), and Velocity (fps)
Data logging: 5-minute intervals (No averaged intervals)

Calibration Performed: Calibration method using 3.50-inch target.

Target Measure:	3.50 in	Meter Read:	3.53 in	10/31/2024 10:25 am

Meter Validation: PASSED

MH1 located at E. Bell Rd. East of N. Hayden Rd.
72" Diameter, Rim to Invert: 144.00 inches
15" PVC pipe, flowing West
One 4" Lateral pipe with no flow.
The pipe condition is intact with a thin layer of scum on the walls.
Scum line of 4.00 inches
Flo-Dar installed pointing upstream in the 15" pipe channel.

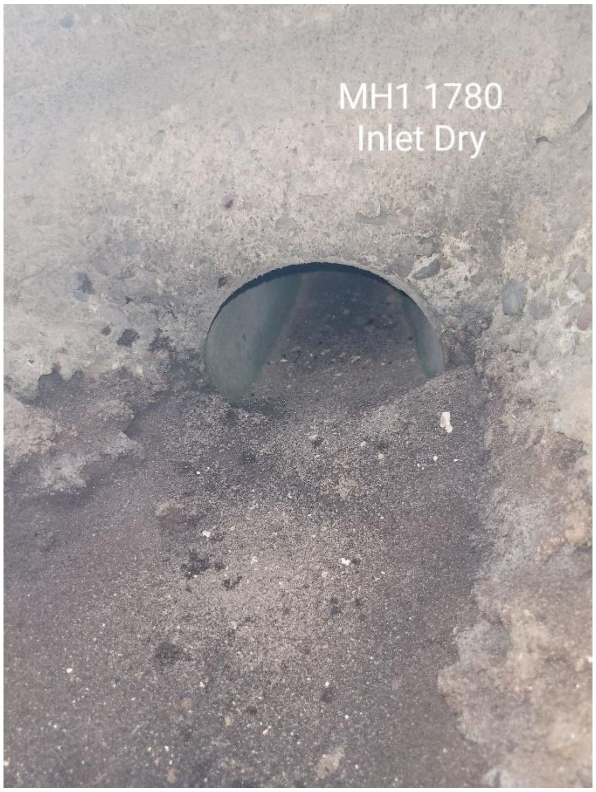
Flow Data is valid having no missing, erroneous, or anomalies with data.

Attached is a MS Excel summary showing level, velocity, and flow logged at 5-minute intervals during the monitoring period.

RDH Environmental Services
Jeff Schulte
Operations Manager
servicemanager@rdh-env.com

SL1780 RDH Flow Study for LGE

Pictures:



SL1780 RDH Flow Study for LGE

Period Summaries:

LGE COS MH1 Period Summary: Flow				
Measures	Value	Unit	Date	Time
Max.	184.97	gpm	Tuesday, November 5, 2024	2:55 PM
Min.	12.58	gpm	Wednesday, November 6, 2024	3:25 AM
Avg.	63.29	gpm		
Total	977,807.00	gal		

LGE COS MH1 Period Summary: Level				
Measures	Value	Unit	Date	Time
Max.	4.06	in	Thursday, October 31, 2024	11:00 AM
Min.	1.51	in	Wednesday, November 6, 2024	4:55 AM
Avg.	2.48	in		

LGE COS MH1 Period Summary: Velocity				
Measures	Value	Unit	Date	Time
Max.	1.60	fps	Tuesday, November 5, 2024	2:50 PM
Min.	0.42	fps	Wednesday, November 6, 2024	3:25 AM
Avg.	0.97	fps		

*Data begins at 10:30 am on October 31st and ends at 4:00 am on November 11th.

SL1780 RDH Flow Study for LGE

Site Map:

CONFINED SPACE ENTRY PERMIT

ALL COPIES OF PERMIT WILL REMAIN AT JOB SITE UNTIL JOB IS COMPLETED

LOCATION/DESCRIPTION OF CONFINED SPACE: MH | AH 1780
DATE: 10/31/24
PURPOSE OF ENTRY: Flow Study Installation
TIME: 9:45am
EXPECTED HAZARDOUS: Gases
EXPIRATION: 12:00pm 10-31-24
COMMUNICATIONS: Hand & Verbal
ENTRY SUPERVISOR: Jordan Astemborski

SPECIAL REQUIREMENTS BEFORE ENTRY:	YES	NO		YES	NO
Lockout De-energize - Test and Verify		✓	Escape Harness Required	✓	
Lines Broken - Capped or Blanked		✓	Tripod Emergency Escape Unit	✓	
Purge - Flush and Vent		✓	Lifelines		✓
Ventilation		✓	Fire Extinguishers		✓
Secure Area (Post and Flag)	✓		Lighting (Explosion proof)	✓	
Breathing Apparatus		✓	Protective Clothing	✓	
Resucitator - Inhalator		✓	Respirator		✓

TEST INTERVAL 15 Min

TEST(S) TO BE TAKEN / ACCEPTABLE ENTRY CONDITIONS
DO NOT ENTER IF PERMISSIBLE ENTRY LEVELS
ARE EXCEEDED

	Permissible Entry Level		DATE / TESTER / TIME AM/PM	10/31 JA 9:55 AM	10/31 JA 10:10 AM	10/31 JA 10:25 AM	M	M	M	M	M	M	M
% of Oxygen	19.5% to 23.5%			20.9	20.9	20.9							
^ of L.F.L.* (Gas/Vapor/Mist)	Less than 10%			0	0	0							
Carbon Monoxide	35 ppm (8 hr.)			0	0	0							
Aromatic Hydrocarbon	1 ppm (8 hr.)			0	0	0							
Hydrogen Sulfide	10 ppm (8 hr.)			0	0	0							
Sulfur Dioxide	2 ppm (8 hr.)			0	0	0							
Ammonia	25 ppm (8 hr.)			0	0	0							

NAME OF GAS TERSTER(S) ______

NOTE: Continuous/periodic tests shall be established before beginning the job.
Any questions pertaining to test requirements should be directed to

TESTING INSTRUMENTS USED	NAME	TYPE	IDENTIFICATION NUMBER
Honeywell	BW Tech	GasAlertMax XT II	XT-XWHM-Y-NA MA215-026608

AUTHORIZED ENTRANTS
Zac Schulte

AUTHORIZED ATTENDANTS
Jordan Astemborski
Eric Gentile

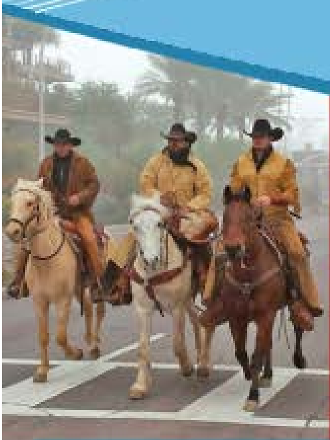
PERMIT AUTHORIZATION

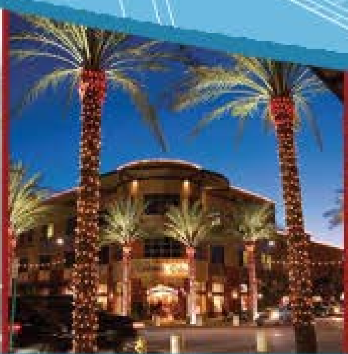
I certify that all actions and conditions necessary for safe entry have been performed

Jordan Astemborski — NAME (Print)
[signature] — Signature
10/31/24 — DATE
10:30am — TIME

IN CASE OF AN EMERGENCY CALL 911

RDH Environmental Services (602) 381-1960





CITY OF
SCOTTSDALE

DESIGN STANDARDS & POLICIES MANUAL

LAND USE	DEMAND (gpd)	DESIGN PEAKING FACTOR
Commercial/Retail	0.5 per sq. ft.	3
Office	0.4 per sq. ft.	3
Restaurant	1.2 per sq. ft.	6
High Density Condominium (Condo)	140 per unit	4.5
Resort Hotel (includes site amenities)	380 per room.	4.5
School: without cafeteria	30 per student	6
School: with cafeteria	50 per student	6
Cultural	0.1 per sq. ft.	3
Clubhouse for Subdivision Golf Course	100 per patron x 2 patrons per du per day	4.5
Fitness Center/ Spa/ Health club	0.8 per sq. ft.	3.5

FIGURE 7-1.2 AVERAGE DAY SEWER DEMAND IN GALLONS PER DAY & PEAKING FACTORS BY LAND USE

HYDRAULIC DESIGN 7-1.404

No public SS lines will be less than 8 inches in diameter unless permission is received in writing from the Water Resources Department.

SS lines shall be designed and constructed to give mean full flow velocities equal to or greater than 2.5 fps, based upon Manning's Formula, using an "n" value of 0.013.

To prevent abrasion and erosion of the pipe material, the maximum velocity will be limited to 10 fps at estimated peak flow. Where velocities exceed this maximum figure, submit a hydraulic analysis along with construction recommendations to the Water Resources Department for consideration. In no case will velocities greater than 15 fps be allowed.

Actual velocities shall be analyzed for minimum, average day and peak day design flow conditions for each reach of pipe.

The SS system shall be designed to achieve uniform flow velocities through consistent slopes. Abrupt changes in slope shall be evaluated for hydraulic jump.

The depth to diameter ratio (d/D) for gravity SS pipes 12 inches in diameter and less shall not exceed 0.65 in the ultimate peak flow condition. This d/D ratio includes an allowance for system infiltration and inflow.

The d/D for gravity drains greater than 12 inches diameter shall not exceed 0.70 for the ultimate peak flow condition. This d/D includes an allowance for system infiltration and inflow.

Measures to mitigate hydrogen sulfide shall be analyzed at manhole drops, abrupt changes in pipe slope or direction and at changes in pipe diameter.

MANHOLES AND CLEAN OUTS 7-1.405

Manholes in city streets shall be located near the center of the inside traffic lane, rather than on or near the line separating traffic lanes. Manholes shall not be in bike trails, equestrian trails, sidewalks, crosswalks or wash crossings. Manholes are required at all

City of Phoenix
Water Services Department

DESIGN STANDARDS MANUAL FOR WATER AND WASTEWATER SYSTEMS

2021

Water Services Department
200 West Washington Street
Phoenix, Arizona 85003-1697
Phone: (602) 495-5601
Fax: (602) 495-5461

in Chapter IV, Section C), are not always adequate to meet water demands. For some projects, a detailed analysis of domestic and fire flow demands may be required to properly define requirements for system design.

1. Water and Sewer Design Flows

The following **Table 8**, *Water and Wastewater Design Flows* shall be used to calculate both water and sewer design flows utilized in the preparation of engineering design reports, plans, and specifications.

Table 8. Water and Wastewater Design Flows.

Land Use	Unit	Water Average Daily Flow/Unit (gal)	Wastewater Average Daily flow/Unit (gal)
Single Family Residential	Dwelling	360	240
Multi-family	Dwelling	240	180
Commercial (retail/mall)	1000 ft^2	125	75
Commercial (office)	1000 ft^2	115	90
Warehousing/Big Box Retail	1000 ft^2	30	25
Industrial	1000 ft^2	65	50
Schools	Student	25	20
Hotel (no restaurant)	Room	140	100
Hotel (with restaurant)	Room	200	150
Resort	Room	300	210
Hospital (all flows)	Bed	500	300
Landscape Water Requirements			
General Landscaping	Acre	4,374	N/A
Public Right of Way or Streetscape	Acre	1,339	N/A
Surface Water	Acre	5,335	N/A

NOTES: The following Italicized notes are for Table 8, Water and Wastewater Design Flows

*Complete design flows are not provided for **industrial and hospital facilities** because case-by-case evaluation is necessary due to varying water demands observed for these use types. Some industrial uses such as data warehouses, food processing, bottling plants, and semi-conductor manufacturing can use more than ten times as much water as compared to warehousing or dry assembly manufacturing with no cooling tower use. Water use in hospitals varies greatly depending upon cooling tower and boiler use, the extent to which the hospital is used as a research and teaching facility, the amount of out-patient versus in-patient services provided, and the types of equipment used. Estimates of anticipated water use and wastewater generation must be produced for each new development or major expansion using projections of demands taking into account the following types of categories:*

- ***Water for cooling towers**: Cooling towers use can make up more than fifty percent of water demand at industrial facilities having large refrigeration units or cooling of servers. In most cases, cooling towers use twenty to forty percent of the water requirements for industrial operations and hospitals.*
- ***Water used as an input for production**: In some manufacturing operations, water is used as an input in the manufacturing process and must be included in demand projections because of the large volumes used. Examples include ice-making, soft-drink or water bottling operations, and food manufacturing such as industrial bakeries.*
- ***Water used in production/activities**: In many manufacturing operations water is used for cooling, cleaning, or other operational activities and must be included in demand projections. Examples include metal forming and finishing, semi-conductor wafer production, and aerospace parts manufacturing. Processes employing newer technologies tend to use less water than older technologies, but estimates must be made on a location and process-specific basis. Some medical facilities are now using the newer medical imaging techniques and sterilization processes that use little or no water, while some medical equipment still requires significant amounts of water.*
- ***Bed to space ratios and mix of services**: Bed to space ratios and services provided in hospitals can vary greatly. These variations depend upon the proportion of space necessary to provide 24/7 nursing care, full linen service, and full food service*

to patients staying overnight. Furthermore, some hospitals are highly specialized and focus on particular types of treatment and/or research while others provide general and emergency services only. Water use on a per-square-foot or per-bed-basis can even vary significantly between different parts of hospitals, so large expansions will require an individual analysis.

2. Water Peak Flow

Peak Flow shall be calculated as 1.7 times the average daily flow.

NOTE: *For clarification, the following example characterizes the calculations performed to determine the design flows and quantities involved in a hypothetical facility.*

EXAMPLE: Hypothetical water demand/flow evaluation (not including fire flows).

ASSUME: A 1000 dwelling unit multi-family development.

CRITERIA: From **Table 8**, *Water and Wastewater Design Flows*.
Average daily flow = 240 gallons per unit per day (gpupd)
Average total daily flow = 1,000 x 240 = 240,000 gallons per day (GPD)
Peak daily flow = 240,000 GPD x 1.7 (peaking factor)
Peak daily flow = 408,000 GPD

3. Sewer Peak Flow

All gravity sewer mains shall be designed for peak flow conditions. Peak flow is calculated as the product of the peaking factor and the average daily flow. The peaking factor should be calculated from Harmon's formula.

Design Flow = Peak Flow = Q Peak = Q avg [1+14/ (4+ $P^{1/2}$)], Where P = Population/1,000

F. WATER AND SEWER MAIN ABANDONMENT METHODS

There are three approved methods of abandoning water and sewer mains in public ROW and easements:

a. Total removal of pipe.
b. Crush pipe in place by mechanical means. This cannot be applied to asbestos cement pipe.
c. Leave pipe in place and fill with low strength grout.

No other methods are acceptable.

G. WATER AND SEWER STUBS OR TAPS AHEAD OF PAVING

City of Phoenix does not allow new stubs or taps ahead of paving unless the property owner can provide a conceptual design report and a site plan demonstrating the appropriate sizing and location of the mains or stubs. This applies to connections such as water/sewer stubs, water/sewer mains and service taps for fire lines and/or domestic use. The request for taps ahead of paving shall be submitted by the developer through a Water and Sewer Technical Appeal.

If the City approves the request for taps ahead of paving, and the size or location changes after the installation due to design changes, or for any other reason, it shall be the property owner's responsibility to abandon any unused infrastructure at the property owner's expense.

H. CROSS CONNECTIONS AND BACKFLOW PREVENTION

1. Cross Connection